

Resume

XIN LIU

Sex: Female

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Education

- M.S. in Harbor, Coastal and Offshore Engineering, Ocean University of China, China, 2022.
- B.S. in Harbor, Waterway and Coastal Engineering, Changsha University of Science and Technology, China, 2015.

Awards and Fellowships

- RESEARDENT for the THERS Make New Standards Program, Apr. 1, 2025 –Sep. 30, 2026 .

Publications

Refereed Journal Articles

1. Xin Liu, Tomoaki Nakamura, Yong-Hwan Cho, Norimi Mizutani. Prototype-scale simulation of key wave parameters and force induced by black tsunamis, *Ocean Engineering*, Volume 353, Part 1, 2026, 124718, DOI: 10.1016/j.oceaneng.2026.124718.
2. Xin Liu, Tomoaki Nakamura, Yong-Hwan Cho, Norimi Mizutani (05 Sep 2025): Three-dimensional simulation of quasi-steady wave forces from black tsunamis: implications for coastal hazard assessment, *Coastal Engineering Journal*, DOI: 10.1080/21664250.2025.2556597.
3. Xin Liu, Tomoaki Nakamura, Yong-Hwan Cho, Norimi Mizutani. Numerical Simulation of Wave-Induced Scour in Front of Vertical and Inclined Breakwaters. *J. Mar. Sci. Eng.* 2024, 12, 2261. DOI: 10.3390/jmse12122261.

Under Review

1. Xin Liu, Tomoaki Nakamura, Yong-Hwan Cho, Norimi Mizutani. Hydraulic and geometric effects on sediment-laden tsunami loading of coastal structures. Submitted to *Coastal Engineering*, 1 July 2026.
2. Yusuf, I. Y. A., Nakamura, T., Xin Liu, Cho, Y.-H., and Mizutani, N. Hydraulic performance of coral reefs for coastal protection: Wave transmission and setup characteristics. Submitted to *Oceans*, 9 June 2026.

3. Xin Liu, Tomoaki Nakamura, Yong-Hwan Cho, Norimi Mizutani. Tsunami-Induced Wave Pressure and Forces: A Review of Clear Water to Black Tsunamis. Submitted to Natural Hazards, 17 Apr. 2026.

Paper Before Doctoral Program

1. Xin Liu, Yang Bo, Liang Bingchen. Numerical simulation of the interaction between suspended sand and moving ore collecting vehicles based on OpenFOAM[J]. Waterways and Ports, 2023, 44(01): 19–30. (In Chinese).
2. Qu Mengxiang, Yang Bo, Liang Bingchen, Xin Liu. Numerical simulation of flow around submarine pipelines in scour terrain based on IDES method. Marine Engineering (02), 97–105. DOI: 10.16483/j.issn.1005-9865.2022.02.010. (In Chinese).
3. Bo Yang, Qin Zhang, Bingchen Liang, Xin Liu. Numerical simulation of the interaction between suspended sediment and moving plate based on the drift-flux model. 2020 Ivannikov Open Conference (ISPRAS), IEEE.

Oral Presentation

1. Xin Liu, Tomoaki Nakamura., Yong-Hwan Cho., and Norimi Mizutani (2026): 3D prototype simulation of black tsunami-induced forces, JSCE Chubu Branch Annual Meeting 2025, JSCE Chubu Branch, II-21, 2 p.
2. Xin Liu, Tomoaki Nakamura., Yong-Hwan Cho., and Norimi Mizutani (2025): Numerical study on wave deformation and force by black tsunamis, 12th International Conference on Asian and Pacific Coasts 2025, p. 135.

Professional Association

- Member of Japan Society of Civil Engineers (JSCE), Mar. 2025-present.